

Probability Theory

Lecture Notes

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1.1 Motivation and Some Problems

Question 1. Sleeping Beauty Problem

A princess is put into a deep sleep by a powerful drug administered by a villain. The drug ensures that she will remain asleep for two years, except for possible brief awakenings determined by the outcome of a fair coin toss.

The villain tosses a fair coin.

- If the coin lands **Heads**, the princess is awakened *once*, briefly, at the end of the **first year**. After this awakening, her memory of having awakened is completely erased, and she is put back to sleep until the end of the second year.
- If the coin lands **Tails**, the princess is not awakened after the first year. Instead, she is awakened *once*, briefly, at the end of the **second year**.

Whenever the princess awakens, she has no information about the current date or the passage of time. Furthermore, if she had awakened previously, all memory of that awakening has been erased. Consequently, upon waking, she cannot distinguish between the two possible awakening scenarios.

Suppose that the princess wakes up and is immediately asked the following question:

“What probability should you assign to the event that the coin landed Heads?”

What should the princess answer?

Solution:

- $\frac{1}{2}$ No information exchanged so probability should be $\frac{1}{2}$, If repeated independently a million times, half chances each.
- $\frac{2}{3}$ There are possibilities of this. Proportion of waking during Heads is $\frac{2}{3}$, and $\dots \frac{1}{3}$
- $\frac{3}{4}$

Villain tosses once, H $\rightarrow$ Villain tosses again, Two Possibilities H and T, so,

- HH $\rightarrow \frac{1}{4}$
- HT $\rightarrow \frac{1}{4}$

- $T \rightarrow \frac{1}{2}$

Important

All the solutions above are wrong. The Above problem has no solution, Because it is ill posed, and this ill posedness is with respect to current Framework.

Question 2. Monty Hall Problem

Monty Hall Problem: Todo: Add figure. There are three doors: one hides a car and the other two hide goats. After you choose one door say A, the host opens another door revealing a goat.

1. Question: Should you switch?
2. Probability car behind C > Probability car behind A.

Solution: Yes, we should change our decision,

Initially, the probability that your chosen door contains the car is $P(\text{chosen door}) = \frac{1}{3}$, while the probability that the car is behind one of the other two doors is $P(\text{other two doors}) = \frac{2}{3}$.

Since the host always reveals a goat, this entire probability concentrates on the remaining unopened door. Therefore, $P(\text{Win by staying}) = \frac{1}{3}$, $P(\text{Win by switching}) = \frac{2}{3}$. Hence, switching gives twice the probability of winning compared to staying.

Summary

- Sleeping Beauty Problem
- Monty Hall Problem

End of Lecture

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